between acidic and alkaline ions); organics (carbon-based industrial chemicals that are toxic and carcinogenic); and a few others, like the presence of metals, ammonia, etc.

There is no water quality index in India that provides a numeric value to capture the overall quality of water like the AQI. The Central Pollution Control Board (CPCB) has identified four parameters to calculate water quality. These are dissolved oxygen, which should be higher than 4 mg per litre; biological oxygen demand (BOD), which should be lower than 3 mg per litre; faecal coliform which should be lower than 2500 most probable number (MPN) per 100 ml; and total coliform, which should be fewer than 5000 MPN per 100 ml. Essentially, it boils down to two parameters: BOD, as higher demand indicates lower availability of dissolved oxygen; and coliform count. Water is evaluated for these aspects. If any value is not met, the water is considered unsatisfactory.

CPCB classifies all unsatisfactory river stretches in five priority categories depending upon BOD.

In May 2019, CPCB announced that the entire water from the Ganga was unfit for direct consumption. Of the 86 live stations monitored, only seven areas were found to have water fit for drinking after disinfection; the rest was not fit for drinking even after disinfection. It is estimated that 70 per cent of surface water in India is unfit for consumption. Another estimate suggests that as much as 40 million litres of wastewater enters rivers, lakes and other water bodies without being treated.

A 2015 CPCB report announced that the number of polluted rivers (whose water is not fit for drinking) in India increased from 121 to 275. In terms of polluted stretches, it found 302 polluted stretches on 275 rivers. This number increased to 351 in 2018. The problem is becoming quite acute in the case of groundwater as well. Besides availability going down, the quality of water, owing to the presence of chemicals and other harmful substances, has also been consistently falling.